Formulas:

1) Range: max - min

2) no of classes:- $1 + 3.3 \log n$ → n → total no.

↓

must always be whole no. 5.4 → 5, 5.5, 5.6 → 6

3) Class size / Interval = $\frac{\text{Range}}{\text{no of classes}}$

23	60	19	32	57	74	52	42	80
36	80	77	87	95	41	65	25	72
55	76	52	10	64	75	78	64	84
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

$n = 60$

Range = 98 - 10 = 88

no of classes = $1 + 3.3 \log 60 = 7$

Interval: $\frac{88}{7} = 12.57 = 13$

grouped frequency distribution

class limits	f	x	fx	fx^2
10 - 22				
23 - 35				
36 - 48				
49 - 61				
62 - 74				
75 - 87				
88 - 100				

How to create class intervals

→ Select the lowest value

⇒ 10 (Lower limit)

10 + 13 → 23 - 1 = 22 (Upper limit)

10 - 22

Second interval 23 (Lower limit)

23 + 13 → 36 - 1 = 35

23 - 35

Qualitative data → describes qualities / characteristics not numbers.

↘ represents categories of things

e.g ↙

Drink	f
coke	10
pepsi	6
Sprite	4

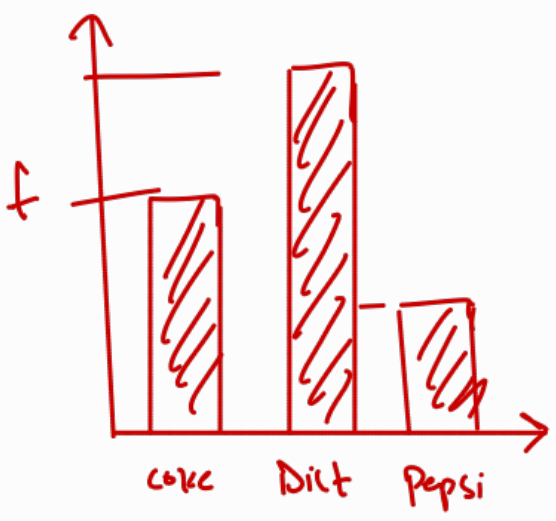
Frequency distribution
→ how often each value appears

Softdrinks	f	Relative f = f/Σf	PRf f/Σf x100	cumulative f
coke	19	0.38		19
Diet	8	0.16		19+8
⋮	5	0.1		5+19+8
	13	0.26		13+5+19+8
	5	0.1		⋮
	Σf: 50	Sum		

Ungroped frequency distribution ↑

Grouped frequency distribution has Class Intervals

Bar chart:-



Both used to represent Qualitative data:

Piechart

$\frac{f}{\Sigma f} = \frac{\text{Obtained quantity}}{\text{Total}} \times 360$



grouped frequency distribution:-

class limits	f	tally	R.F	C.F	PRF	x (midpoint)	fx	fx^2
10 - 22	3	III	3/60	3	5%	$\frac{10+21}{2}$	48	768
23 - 35	4	IIII	⋮	7	6.67%	29	116	⋮
36 - 48	5	𝍸	⋮	⋮	⋮	⋮	⋮	⋮
49 - 61	8	𝍸 III	⋮	⋮	⋮	⋮	⋮	⋮
62 - 74	14	𝍸 𝍸 III	⋮	⋮	⋮	⋮	⋮	⋮
75 - 87	20	𝍸 𝍸 𝍸 𝍸	⋮	⋮	⋮	⋮	⋮	⋮
88 - 100	6	𝍸 I	⋮	⋮	⋮	⋮	⋮	⋮

$\Sigma f = 60$

Mean (grouped data) $\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{3480}{60} = 65.83$

Mean (ungrouped data) $\bar{x} = \frac{\Sigma x}{n}$

Standard deviation:-

$$S = \sqrt{\frac{1}{n-1}\left[\Sigma fx^2 - \frac{(\Sigma fx)^2}{n}\right]}$$

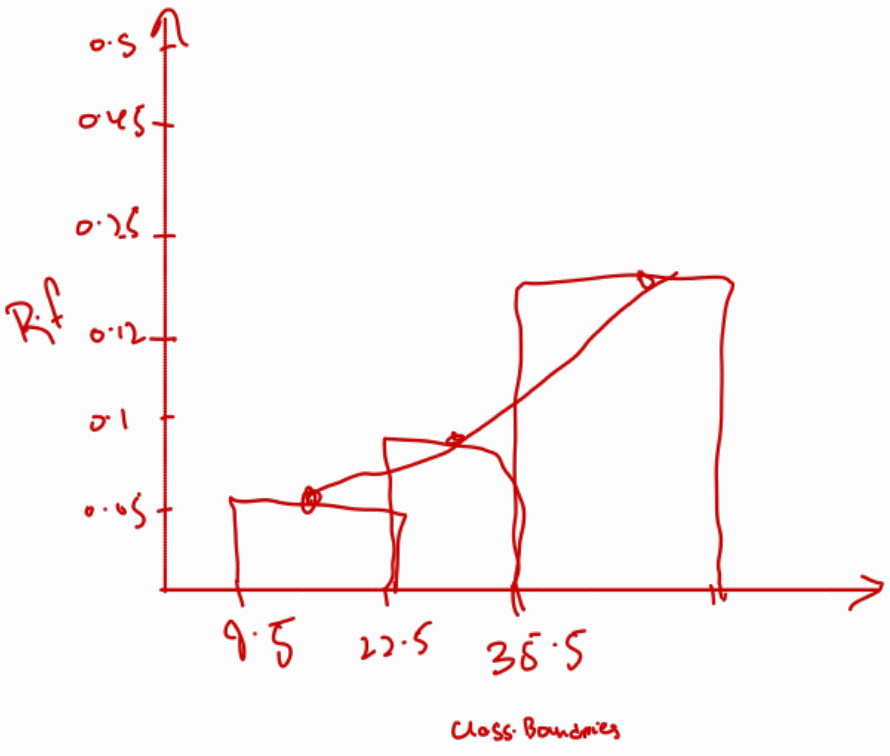
$n \rightarrow \Sigma f$

Variance $= (S.D)^2$

(Relative frequency histogram)

R.f :- $\frac{\text{frequency of class}}{\text{Total f}}$

x-axis → class intervals
y-axis → R.f



gap b/w class

$\frac{23-22}{2} = 0.5$

Subtract 0.5 from lower class and add 0.5 to upper class.

Another SD formula-

$$S.D = \sqrt{\frac{\leq f(x-\bar{x})^2}{n-1}}$$

$(x-\bar{x})^2$ | $f(x-\bar{x})^2$

$$S.D = \sqrt{\frac{1}{n-1}\left[\Sigma fx^2 - \frac{(\Sigma fx)^2}{n}\right]}$$

Σf → grouped data

$$S.D = \sqrt{\frac{1}{n-1}\left[\Sigma x^2 - \frac{(x)^2}{n}\right]}$$

→ ungrouped data

Q 5point Summary of the data and make Boxplot and See outliers

3450, 3840, 3550, 3925, 3650, 3520, 3480, 3490, 3355, 3310, 3490, 3730.

2) Sort in ASC.

Five point

min:- 3310

$i = \frac{p}{100} \times n$ → percentile to calculate

integer i → Average i and i+1

non-int → Round up i

$i \rightarrow$ tells position.

$Q_1 = \frac{25}{100} \times 12$

$i = 3$

$\frac{3450 + 3490}{2} = 3465$

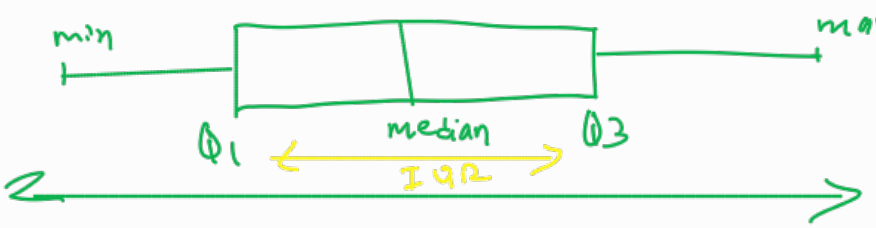
$Q_2 = \frac{50}{100} \times 12 = 6$

$\frac{3490 + 3520}{2}$

$= 3505$

$Q_3 = \frac{75}{100} \times 12 = 9$

⋮



Box-plot

$IQR = Q_3 - Q_1$

outliers formula

L Limit $= Q_1 - 1.5(IQR)$

U Limit $= Q_3 + 1.5(IQR)$

L.L $= 3262.5$ Anything Less than that is outlier

85 percentile

$= \frac{85}{100} \times 12$

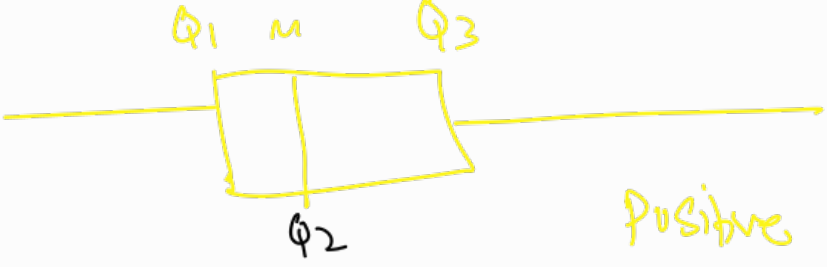
$= 10.2 \approx 11$

3730

D_3 means 30th percentile

(11)

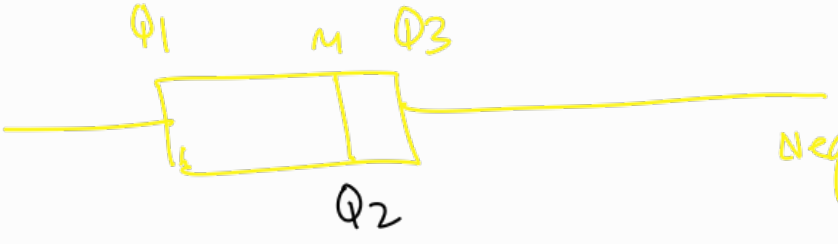
$P_{80} \rightarrow$ 80th percentile



$Q_2 - Q_1$

Positive Skewed

$Q_3 - Q_2 > Q_2 - Q_1$



Neg Skewed

$Q_3 - Q_2$

$Q_3 - Q_2 < Q_2 - Q_1$

*

$Q_1 = 345 \rightarrow$ 25% of graduates earn less or equal to 345.

z^2 method to calculate Outliers:
Answer may be different as compared to 5 point Summary

$$Z = \frac{X - \text{mean}}{Sd} \rightarrow (\text{original point})$$

Mean - 3510
SD = 165.683

Z	X
-0.5433	3450
0.0604	3550
0.6640	3650
-0.3621	3480
-1.1168	3355
-1.3684	3310
-0.3018	3490
1.1469	3730
0	3540
2.3241	3925

given in Q

$Z \leq / \geq$ value

Boundary itself is not a outlier.

anything exceeding / decreasing this value will be a outlier.

there is no outliers according to the Z-Score method

Relationship b/w mean, Median, mode, works only if data is skewed Right/left.

Mode :- 3 Median − 2 Mean

Skewness

Pearson Skewness :- $\frac{\text{Mean} - \text{Mode}}{S.D}$

Sk > 0 Right skewed
Sk < 0 left skewed.

Central tendency:- middle value

↳ main measures are mean, median, mode

Spread/Variability:-

↳ main measure are IQR, Range, Standard deviation

For Box/Dot plot

Larger IQR means Inconsistent data all over the place.

Smaller IQR means consistent data.

→ median.

(Trimmed mean) → sorted

1) make sure your data is sorted first.

no of values to remove = n × p

↓ → percentage

Benifit:- ↓

chops off the outliers while your making mean

5 12 13 15 16 17 18 19 20 95

20% trim

10 × 20/100 = (2)

$$\frac{13+15+16+17+18+19}{6} = (16.5)$$

cross-tab | Joined frequency table | tells two diff things at once

Size	Premium	Regular	Diesel	
compact	1	3	0	4 → total frequencies
Midsize	0	3	1	4
Large	2	0	0	2
Total	3	6	1	10

marginal totals

Qualitative var → The labels

Quantitative var → The measurement. | must be equal dis

Type of Measurement:-

Nominal	Ordinal	Interval	Ratio
Naming / no order	order matters	Dis b/w no is equal	cannot go below 0.
Labels only	↓ Data can be arranged in a specific order	e.g Temp can be -10°C, you can go below 0.	e.g having -5 apples, -10kg weight.
↓ Data cannot be arranged in a specific order	↓ Qualitative	no such thing as twice as much, thrice etc	True zero ↓ meaning its just like a scale no scale
		No true zero meaning if anything is 0 it doesn't mean it doesn't exist. zero	

Z-Score $Z = \frac{x - \bar{x}}{\sigma}$

x = value
$\bar{x}$ = mean
σ = S.D

$-3 \leftarrow |z| \rightarrow 3$

if any value of z is less than -3 → Outlier
or if greater than 3 → Outlier

write Z-Scores to 4 dp.

CV → measure of variability
Spread of data as a % of mean

$CV = \frac{SD}{\bar{x}} \times 100$

Used to compare 2 groups

Low CV → consistent / Data packed together

High CV → less (widely scattered / less consistency)
↓ greater variation

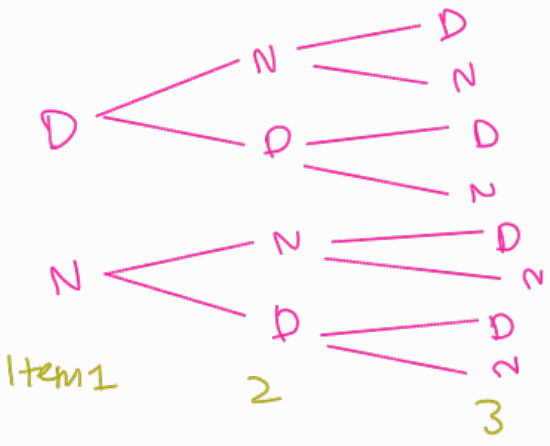
Many media mode → positively skewed

Mean < median / mode → negatively skewed

Tree diagram for listing Sample Space

D → Defective, N → non-defective

∴ three items are selected at Random



DND, DNN, DDD, DDN,
NND, NNN, NDD, NDN
↓ set of possible states

$S = \{x \mid x \text{ is a city} \Rightarrow 1000 \text{ ppl}\}$

x is ↓ such that x is a city with ppl of over 1000

Tailing

Subset of a Sample Space gives idea of Event (capital letter)

DND → 1 state → Single event

DND, NND, NNN → more than 1 / multiple → compound event

counting of Sample points

→ Rule of multiplication, permutation, combination → order doesn't matter.

objects : Spaces

n_{Cr}

↓ m+n

(event happening one after another) Chaining.

↓ objects ! = Spaces order matters

objects: Spaces → same items in same no of Spots → factorial

2 objects = 2!

class: 30, we need 3 student to give First, Second, this position.

$^{n}P_{r}$ → $\frac{n!}{(n-r)!}$

$^{30}P_{3}$

words like arrangement indicate we will use permutation

$^{n}P_{r} = {}^{n}C_{r} \times r!$

$^{n}P_{r}$ ↓ chose things with order

$^{n}C_{r}$ ↓ chose without order

$r!$ ↓ Arrange them

Mutually exclusive → can two events happen in the same time

if no → mutually exclusive
if yes → not mutually exclusive

Dependent events : events that effect each other.

→ pouring rain outside and chance of getting into a accident.

$P(A \cap B) = P(A) \times P(B)$ in independent

$P(A \cap B) = P(A) \times P(B|A)$ in dependent

Bayes theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

↓

Question asks reverse condition

e.g if x is supplied and the prob of being defective is 15%. what prob will be if computer is already defective, which part it is made of.

Random Var → assigns a real val to all possible outcomes in a Sample Space

Σ 1, 2, 3, 4, 5, 6
↓ ↓ ↓ ↓ ↓ ↓
0 1 2 3 4 5 → R.V

e.g

Sample Space	y
RR	1
RB	1
BR	2
BB	0

S.S is called discrete if it has
countable outcomes (infinite)
finite outcomes OR

e.g Σ 1, 2, 3, 4, 5, 6 3 OR Σ FF, NFF, FFF, NFJF .....
outcomes → 6 ↓ ↓ ↓ ↓
1 2 3 4

continuous S.S :: → Infinite + vals are in pdnk.

B/w 1 and 2
1.5
1.55
1.555
1.5555

Probability function::

(x, f(x))
↓ ↳ its probability
val

(Discrete probability distribution::) → for discrete Ran

1) f(x) > 0 Prob cannot be neg
2) Σ f(x) = 1

Probability func ↳ list of values and there probabilities

(Mean of Random Var)

Expected value = avg val you expect in long run.

$$E(x) = \sum (x \cdot f(x))$$

multiply every val by its probability and add togather.

7 $\quad$ 4 → good $\quad$ 3 → defective $\quad$ | 3 → Sample is taken

X = 0 $\quad \frac{3}{7} \times \frac{2}{6} \times \frac{1}{5} = \frac{6}{210}$ (All defective)

X = 1 $\quad \frac{4}{7} \times \frac{3}{6} \times \frac{2}{5} = \frac{4}{35}$ (only 1 good)

$$\frac{3}{7} \times \frac{4}{6} \times \frac{2}{5} = \frac{4}{35}$$

$$\frac{3}{7} \times \frac{2}{6} \times \frac{4}{5} = \frac{4}{35}$$

$$\frac{4}{35} + \frac{4}{35} + \frac{4}{35} = \frac{12}{35}$$

X = 2 $\quad \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5} = \frac{6}{35}$ 2 good

$$\frac{4}{7} \times \frac{3}{6} \times \frac{3}{6} = \frac{6}{35}$$

$$\frac{3}{7} \times \frac{4}{6} \times \frac{3}{5} = \frac{6}{35}$$

→ $1\frac{6}{35}$

X	P(X)
0	1/35
1	12/35
2	18/35

ask Khalsa

multiple ways for same event → add these probs.

$$E[g(x)] = \Sigma\ g(x) \times f(x)$$

$$\sigma^2 = \Sigma\left((x-\mu)^2 \cdot f(x)\right)$$

value · mean · prob

x	P(x)
1	0.5
3	0.5

$$\mu = (1 \times 0.5) + (3 \times 0.5) = 2$$

$$(1-2)^2 \times 0.5 + (3-2)^2 \times 0.5 = 1$$

(cumulative distribution function) → discrete random var

$F(x) = P(X \leq x)$ adding all probabilities together

$F(x) = f(0) + f(1) + f(2) + f(n) = 1$ final sum

x	f(x)
0	0.2
1	0.5
2	0.3

→

x	F(x)
0	0.2
1	0.7
2	1

if CDF is given and we want to find the exact prob of $X=1$ then

exact = that value − previous

$$P(X=1) = F(1) - F(0) = 0.7 - 0.2 = 0.5$$

Plot

→ continuous Random var

(Probability density function)

1) $f(x) \geq 0 \quad x \in \mathbb{R}$

2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

e.g $f(x) = \begin{cases} x^2/3 & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$

(a) verify $f(x)$ is a density function

↳ look at both functions and check if $f(x) \geq 0$ → must always be + or 0.

✓ pass

(b) find $P(0 < x \leq 1)$

$$\int_0^1 \frac{x^2}{3}\,dx = \left| \frac{x^3}{3 \times 3} \right|_0^1 = \frac{1}{9}$$ ✓

cumulative distribution function → (continuous Random variable)

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

always start integ limits where the fun starts

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$P(0 < x \leq 1)$

$$F(x) = \int_{-1}^{x} \frac{x^2}{3}\,dx \rightarrow \left| \frac{x^3}{3 \times 3} \right|_{-1}^{x}$$

$$= \frac{x^3 + 1}{9}$$

e.g.

$$f(y) = \begin{cases} \frac{5}{8b} & \frac{2}{5}b \leq y \leq 2b \\ 0 & \text{elsewhere} \end{cases}$$

Find $F(y)$ $= \int_{\frac{2}{5}b}^{y} \frac{5}{8b}\,dy = \frac{5t}{8b}\Big|_{2b/5}^{y} = \frac{5y}{8b} - \frac{1}{4}$

Joint probability distribution

$f(x,y) \geq 0$ → two probability happening

$\sum_x \sum_y f(x,y) = 1$

$P(X=x, Y=y) = f(x,y)$

P=2 3-Blue 2-Red 3-green

X → no of Blue pens

Y → no of Red pens

(a) (0,1) (1,1) (1,0) (0,2) (2,0)

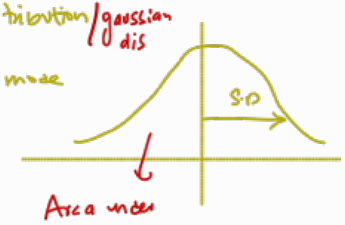
(0,0)

P(x,y)	y = 0	100	200
100	.20	.10	.20
x 250	.05	.15	.30

P(100, 0) = 0.20

Ch: 6 normal Distribution / gaussian dis

mean = median = mode



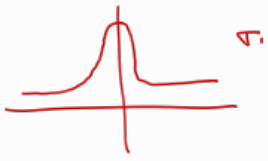
Area under the curve is 1 since represent probability

normal function..

$$n(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

variable, mean, S.D

$\sigma \Rightarrow$ spread of curve



$\sigma_2 > \sigma_1$



Mode $\Rightarrow$ highest point (peaks) of the curve (x value)

mode: μ_1

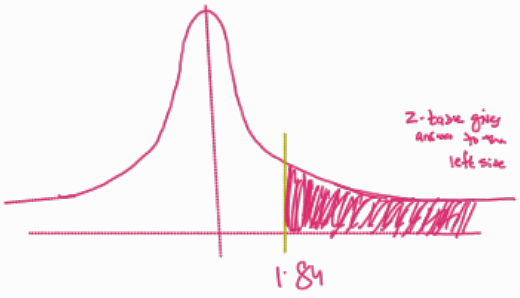
$E(X) = \mu$

(Area under normal curve)

1) Data originally in x, tranform it in $z = \frac{x-\mu}{\sigma}$ because table only understands z.

2) check table for val of prob corresponding to z values.

Standard normal distribution ($\mu = 0$, $\sigma = 1$)



1.84 → 1.8 + only the last digit 0.04

2.73 → 2.7 + 0.03

-1.97 → -1.90 + 0.07

e.g 6.3/1

→ Reverse finding value

$P(z > k) = 0.3015$

Area to left $= 1 - 0.3015$

$= 0.6985$ → this is Area note that

↓ from table

$k = 0.52$

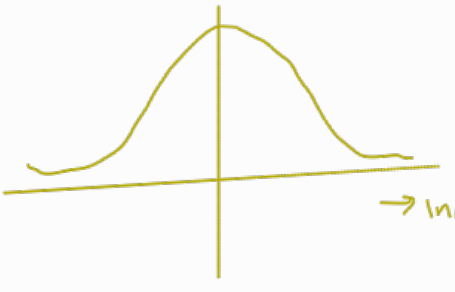
find area under the curve:



finding area of 0.86 gives left side

are then finding area of -1.97 minus Both.

$0.86 - (-1.97)$



find 6th decile? → means 60% of area is to left

find 85 percentile → 85% area to the left always.

Ch #9

Influential Statistics:: to draw conclusions about population on the Basis of Sample) | using info from a small group to make conclusions about large group.

1- estimation of parameters (population characteristics estimation by sample)

(a) point est
↳ interval est / gives Range of values

2- Hypothesis testing of parameters (population parameters assumptions test on the Basis of Sample)

Both formulas has α :: ↓ margin of error / level of significance / type 1 error.

α standard value is 5%.
confidence Level is 95%.

note
S.D known → z-test (Sences normal)
S.D unknown → t-test
↓
2 para : degree of freedom (df)
↓
v = n-1
↓
Sample size

greater df means more Symmetric Shape

(C.I)

Single Sample:: Estimating the mean formula → means using a single sample to estimate the mean of an entire population.

S → Sample S.D
σ → population S.D

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

↳ confidence interval

$\bar{x}$ → mean
μ → what we want to estimate (population mean)
σ → popln S.D
n = size
$z_{\alpha/2}$ = z-value based on confidence level → α = 1 − confidence Level
$\frac{\sigma}{\sqrt{n}}$ = error

if C.I = 90% → 0.9 → (0.1) then divide by 2 and check corresponding z value

note: this gives us area.

one-sided upper Bound

$\bar{x} + Z_\alpha \sigma/\sqrt{n}$

$\bar{x} + Z_\alpha \sigma/\sqrt{n}$

e.g $\alpha_{0.025} \rightarrow 1-0.025 = 0.9750$ → 1.96

↓

Area, now

Look at the corresponding Z-values.

$Var = (S.D)^2$ e.g 9.2 e.g 9.5, 9.5

((-2) calculating population mean when population S.D (σ) is not known (t-test)

$$\bar{X} - t_{(\alpha/2, n-1)} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{(\alpha/2, n-1)} \frac{S}{\sqrt{n}}$$

upper Bound

$\bar{X} + t_{(\alpha, n-1)} \frac{S}{\sqrt{n}}$

↓

Sample size

↓

confidence interval

$n = 500$

$\bar{X} = 501$

$S = 112$

We are 95% confident that answer lies in Range

Conclusion

if n is very Large ($n >= 30$) use z-test even if S.D is not known

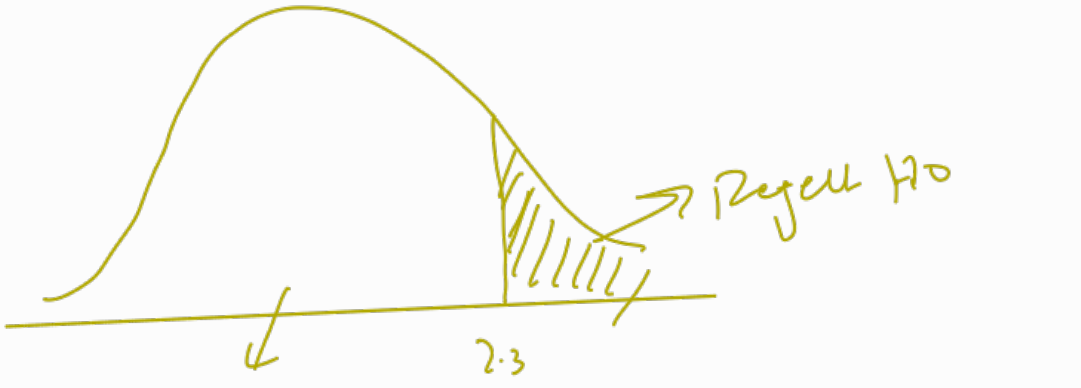
(Hypothesis-testing) → making a decision

H_0 (null hypothesis)
not Significant / no effect

H_1 (Alternate hypothesis) → Significant, has effect
we are trying to prove this.

Type I error: (α)
↳ happens when H_0 is true but we reject it.

Type II error (β) happens when H_0 is false and we fail to Reject it.



(Steps of hypothesis-testing)

1) State the null / alternate hypothesis
2) Level of Significance (α) → not given 5%.
3) establish CR.
4) Reject / Accept H_0

One-tail

$H_0: \theta = \theta_0$

$H_1: \theta > \theta_0$

Right tail

$H_0: \theta = \theta_0$

$H_1: \theta < \theta_0$

left

Two-tail:

$H_0: \theta = \theta_0$

$H_1: \theta \neq \theta_0$

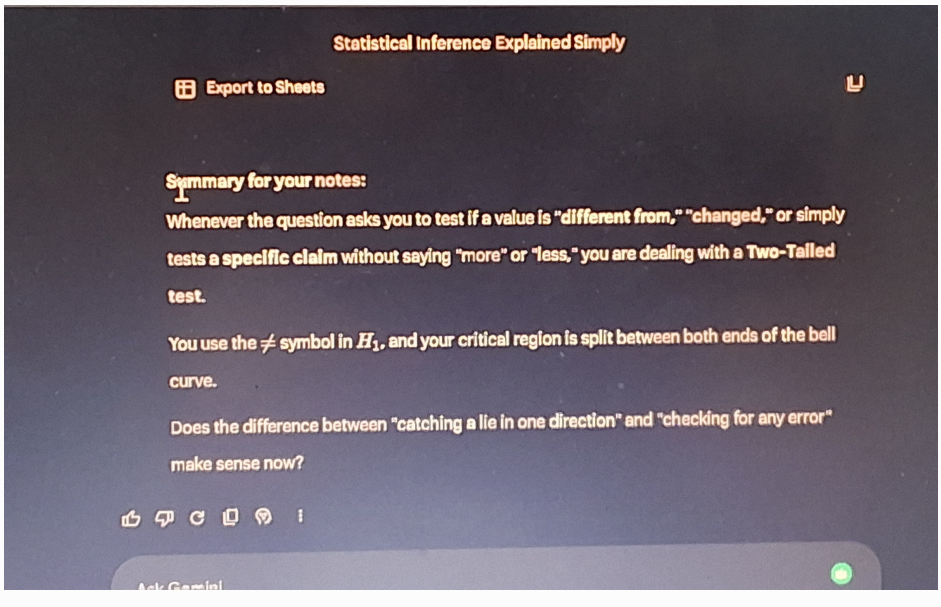
$H_0 \rightarrow$ normal

$H_1 \rightarrow$ write what you are trying to claim / prove

Note: in $H_0 \rightarrow$ always write $(=), \geq, \leq$

$H_1 \neq, <, >$

if H_1 has $\neq$ then it is 2-tailed



In hypothesis testing we are always looking for the conflict

Testing hypothesis about Single mean:

- z-test (population variance is known) / S.D

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- (t-test) population variance is unknown / we have to work with Sample S.d / Variance

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

($\bar{x}$: Sample mean; μ: population mean)

After finding values from these formulas we compare critical values

Testing for reduction / left tailed

Reject H_0 if

$z \leq -z_\alpha$ $\qquad$ $t \leq -t_{\alpha, v}$

Testing for increase / Right tailed

Reject H_0 if

$z \geq z_\alpha$ $\qquad$ $t \geq t_{\alpha, v}$

Two tailed

$|t| \geq t_{\alpha/2, v}$

$|z| \geq z_{\alpha/2}$

Regression: way to use data from past to make educated guesses abt the future

1) find dependent variable (y) and Independent variable (x)
   - dependent: unknown
   - Independent: known quantity

2) estimated Regression eq. is $\hat{y} = a + bx$

$$b = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

multiply x and y then add

square every x value first then add

$$a = \bar{y} - b\bar{x}$$

↙ y-intercept

3) unexplained variation and total variation:

Total variation: overall variation / spread

unexplained variation: sum of all leftover mistakes or Regression formula doesn't account

→ Total variation: $\Sigma(y - \bar{y})^2$ → Subtract first then take square, then sum them

(Total Sum of Squares)

↓ actual point

↓ mean

→ unexplained SSE $\Sigma(y - \hat{y})^2$

coefficient of determination: / Percentage of accuracy

$r^2 = 1 - \frac{\text{unexplained}}{\text{Total}}$

→ Total - unexplained = explained

(SSR)

Explained formula $= \Sigma(\hat{y} - \bar{y})^2$

$r = \sqrt{\ldots}$

↓ less strength of relationship

-1 or +1 v.v strong

multiple Regression..

$$y = a + b_1 x_1 + b_2 x_2 \rightarrow b_3 x_3$$

(b_1: ind variables; b_2: weights)

a → y. int

2 or more inputs to guess the same output.

Multiple Regression gives a belt to guess then on

(Hypothesis testing in Regression) → making a claim so that we can test it.

ANOVA approach

↓ testing on Slope

1) Ho (null hypothesis) → nothing is happening $b_1 = 0$.

H_1 : Something is happening, slope $\neq 0$

if we reject Ho → I have evidence that x significantly predicts y.

if we fail to reject Ho → Slope is close to 0.

3) α: 5% threshold

(Building Anova table) → fixed

F(α, df1, df2) → df1: df of explained; df2: df of unexplained

Sources of var (SOV)	df	SS (Sum of Square)	MS (mean square)	F-Stat	P-val
Regression	1 (fixed)	Explained	MSR = SSR / 1	F = MSR / MSE (explained / unexplained)	
error	(n-v) - 1	Unexplained	MSE = SSE / n-2		
Total	n - 1 (↓ Sample Size)	Total			

Look at Anova table to find

$$R^2 = \frac{(\hat{y} - \bar{y})^2}{(y - \bar{y})^2}$$ → coefficient of determination

$r = -\sqrt{R^2}$

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$ > Same thing

↓ correlation coefficient

not In practice 0.7 or 0.8 is considered strong relationship

$\alpha = 0.05$

0.6 — moderate

$\Sigma(\hat{y} - \bar{y})^2$ → SSR / explained

$\Sigma(y - \hat{y})^2$ → SS Residual / unexplained

$\Sigma(y - \bar{y})^2$ → Total

Decision Rule: if p-value $\leq \alpha$ then reject H_0 : model is Significant else donot Reject H_0 model is insignificant.

OR

$F_{Ratio} \geq F(\alpha, df_1 (Regression), df_2 (Error))$ Reject H_0

if p-value is not given

Hypothesis testing about Regression coefficients | t-test

Anova table (F-test) → gives overall model Significance VS t-test gives Individual parameter Significance | Slope Significance

$F = t^2$ → Relationship

(t-test)

1) $H_0 : \beta_1 = \beta_s$ (Slope is equal to a specific val) → 0 (x doesn't influence y)

$H_1 : \beta_1 \neq \beta_s$ (Slope is not equal to a Specific val)

↓

x influences y.

2) $\alpha/2$

find $t(\alpha/2, n-2)$ e.g. $(t_{0.025}, 18)$ → find val from t-table

3) now compute

$$t = \frac{b - \beta \to 0}{SE(b)}$$

b → slope

where

$$SE(b) = \sqrt{\frac{\Sigma (y - \hat{y})^2}{n-2}} \quad (n-2 \rightarrow \text{degrees of freedom})$$

$$\frac{}{\sqrt{\Sigma (x - \bar{x})^2}}$$

4) Reject H_0 if $t < -t(\alpha/2, n-2)$ or

means Model has Significance / Slope exists $t > t(\alpha/2, n-2)$



$$b = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

(sum of x-y cross product / sum of x square)

$$t = \frac{b - \beta}{S_{E(b)}}$$ (b → slope, β → specific value)

where

$$S_{E(b)} = \frac{\sqrt{\frac{\Sigma(y-\hat{y})^2}{n-2}}}{\sqrt{\Sigma(x-\bar{x})^2}} = \sqrt{\frac{\Sigma y^2 - b_0\Sigma y - b_1\Sigma xy}{n-2}}$$

(b_0 → y-intercept)

$$\sqrt{\Sigma(x-\bar{x})^2} \rightarrow x^2 - \frac{(\Sigma x)^2}{n} = \sqrt{\ldots}$$

Testing of hypothesis → ANOVA Simple linear Regression (F-test) → overall Significance

// Slope → t-test → Significance b/w x and y

// correlation coefficient → t-test → a Significant result proves x and y move together

Hypothesis of correlation coefficient → measures significance of Relationships, its strength and direction. / checks whether Relationship is real or not.

1)

Ho $\rho = 0$ (no linear correlation) → X tells you nothing obj y

H1 $\rho \neq 0$ (linear correlation exists)

2) $\alpha = .05$

3) $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$

$n-2$ is d.f

$r = \sqrt{r^2}$

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

4) Reject Ho if $t > t(\alpha/2, n-2)$ pos or

$t < -t(\alpha/2, n-2)$

↓

concludes

Model is statistically Significant.

↓

neg Relationship

$t \geq t(\alpha/2, n-2)$

$t \leq -t(\alpha/2, n-2)$

↓

2 tail

Tails:: → directions we look at for evidence to reject Null Hypothesis (H0)

Two-tailed

≠0 (Test for significances use $\alpha/2$

H_0 Reject only if

$t \geq t(\alpha/2, n-2)$

or

$t \leq -t(\alpha/2, n-2)$

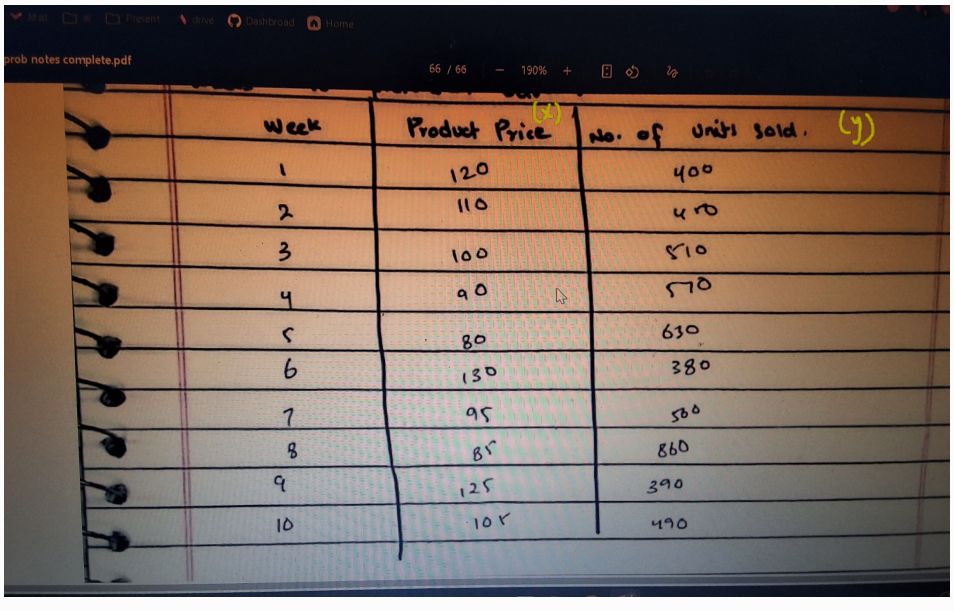
one-tailed

H_0

> Reject only if $t \geq t_\alpha$ use α

< Reject only if $t \leq -t_\alpha$ use α

H_0



Week	Product Price (x)	No. of units sold. (y)
1	120	400
2	110	470
3	100	510
4	90	570
5	80	630
6	130	380
7	95	580
8	85	860
9	125	390
10	105	490

Test whether there is a -ve correlation b/w prices and units sold.

H_0: $\rho \geq 0$ → The opposite of what we want to test.

H_1: $\rho < 0$ → Always the thing we wat to test: in this case negative correlation.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$P < 0 \quad : \quad t \leq -(t_{\alpha}, df)$

↓

Because we want neg values

hence $\leq$

Median of grouped data $= l + h\left(\frac{n}{\Sigma} - cf\right)$

- l → class with median → use i formula
- cf → cumulative frequency before median
- h → frequency of median class
- l → l.l of median class

if gap b/w classes exist take them Avg

coefficient of variation

$CV = \frac{S}{\bar{x}} \times 100\%$

S → Sample. sd

To find Relationship b/w variables use r formula

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$R^2 = 1 - \frac{\text{explained}}{\text{Total}}$ → tells goodness of fit

↓

coefficient of determination

Correlation coefficient

$$r = \pm \sqrt{R^2}$$

$$\hat{y} = a + bx$$

$$b = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

multiply x and y then add

Square every x value first then add

$\Sigma(\hat{y} - \bar{y})^2$ → explained SSR

$\Sigma(y - \hat{y})^2$ → SSE

$\hat{y} = 37.1217 + 0.5176x$

Possion distribution

Uses
→ no fixed no of trails
→ gives average rate over a time window

$$\frac{\mu^x e^{-\mu}}{x!} \rightarrow \text{mean}$$

3 : 10
4 5

15 = 10n

$n = 1.5$

Atmost in Possion

$P(x \leq 5) = P(x=0) + P(x=1) + P(x=2) \cdots P(x=5)$

$P(x < 5) = P(x=0) + P(x=1) + P(x=2) \cdots P(x=4)$

$P(x \geq 3)$ atleast in possion

meaning exactly 3 or 4 or million.

$$1 - [P(x=0) + P(x=1) + P(x=2)]$$

$$\binom{n}{r} = nCr = \frac{n!}{(n-r)!\,r!}$$

Binomial

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

p → success, q → fail

$$\text{cov}(x,y) = E(xy) - E(x)E(y)$$

$E(xy)$: xy × real prob

$E(x)E(y)$: multiply each x value by its column total; multiply each y value by its row total